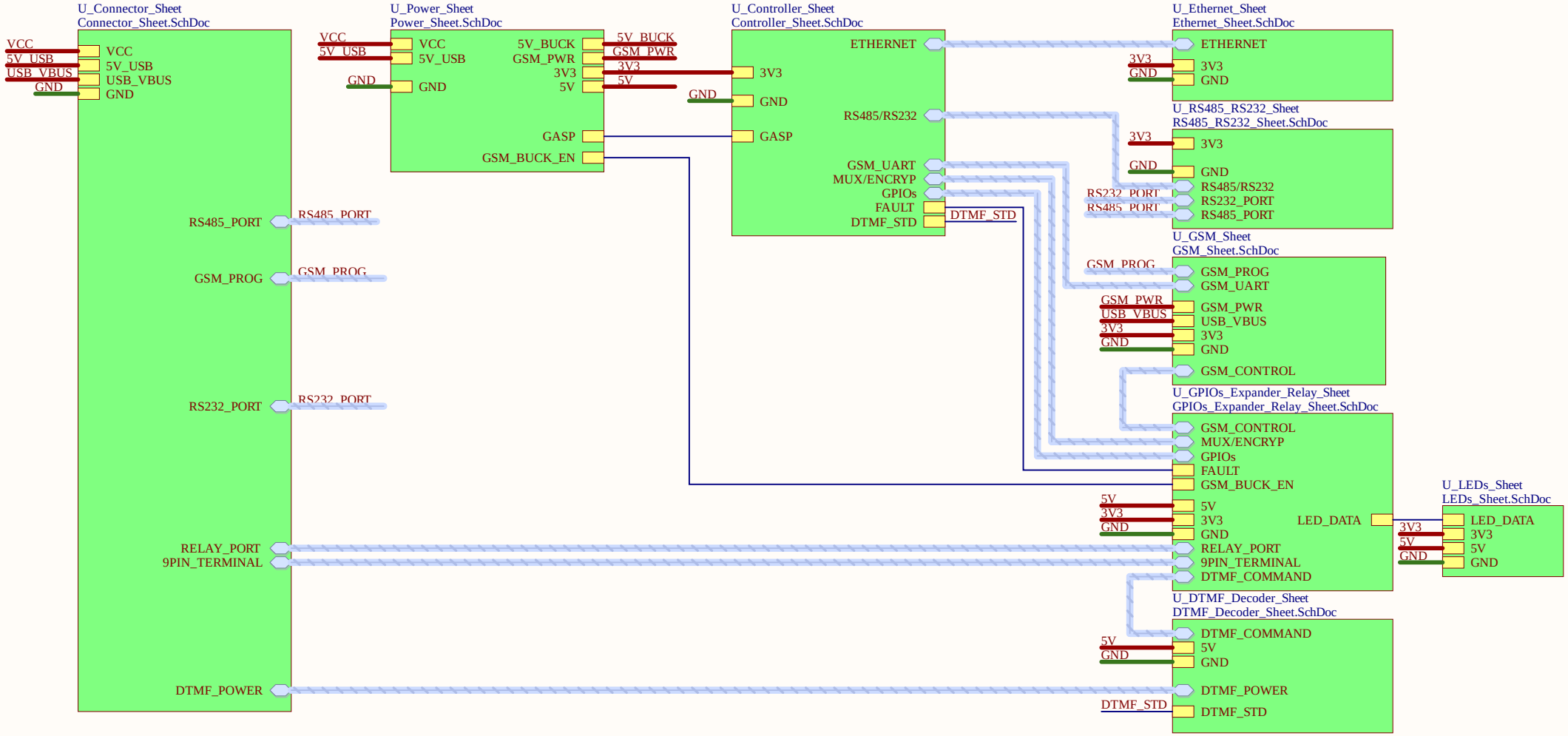
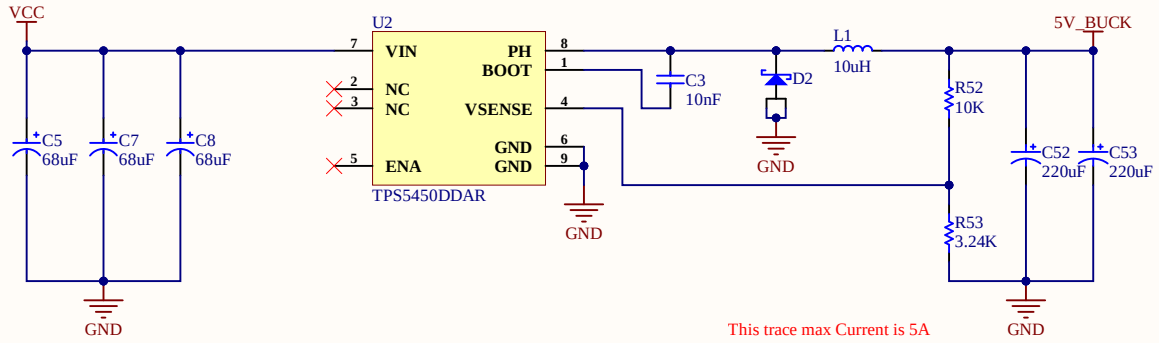


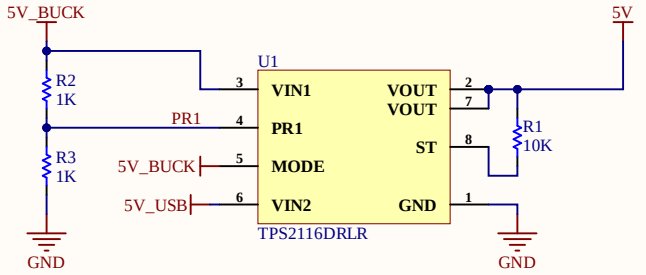
eSpark Consulting Group

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File: ESP32IOTProduct.PrjPcb	
Date: 08/05/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 10

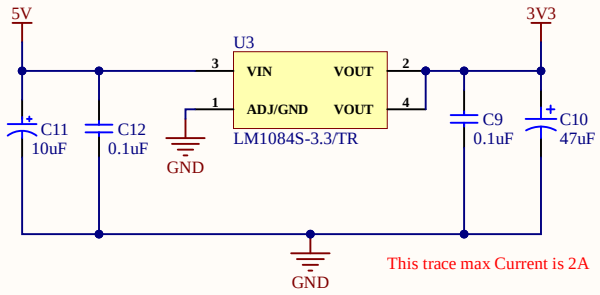
5V BUCK REGULATOR



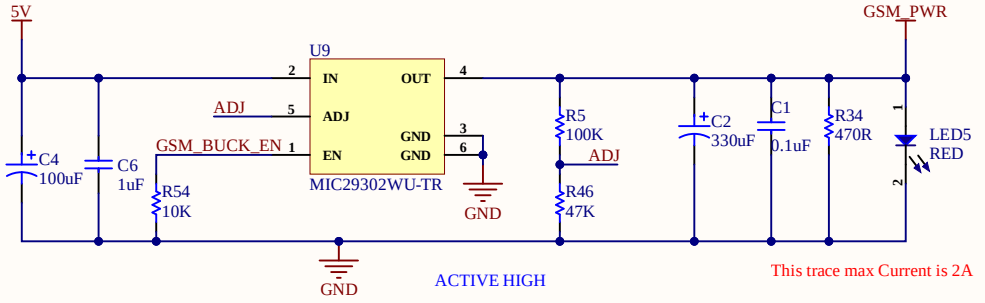
ORING CIRCUIT



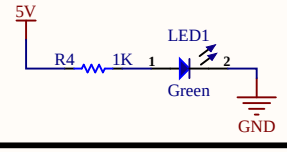
LDO 3.3V

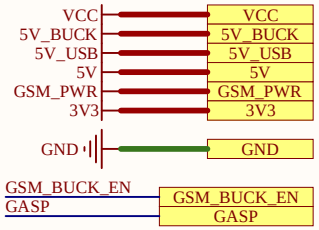


BUCK FOR GSM POWER



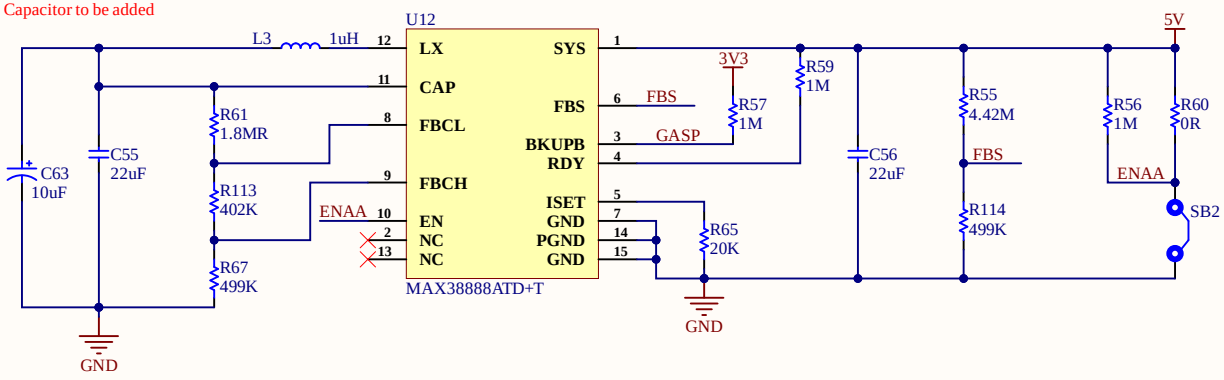
POWER LED





DYING GASP IC

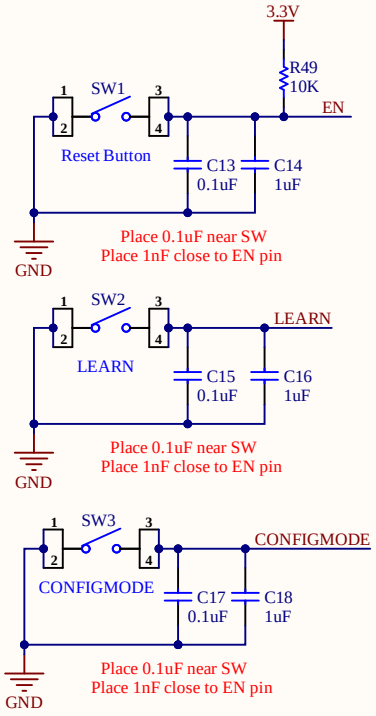
100F Capacitor to be added



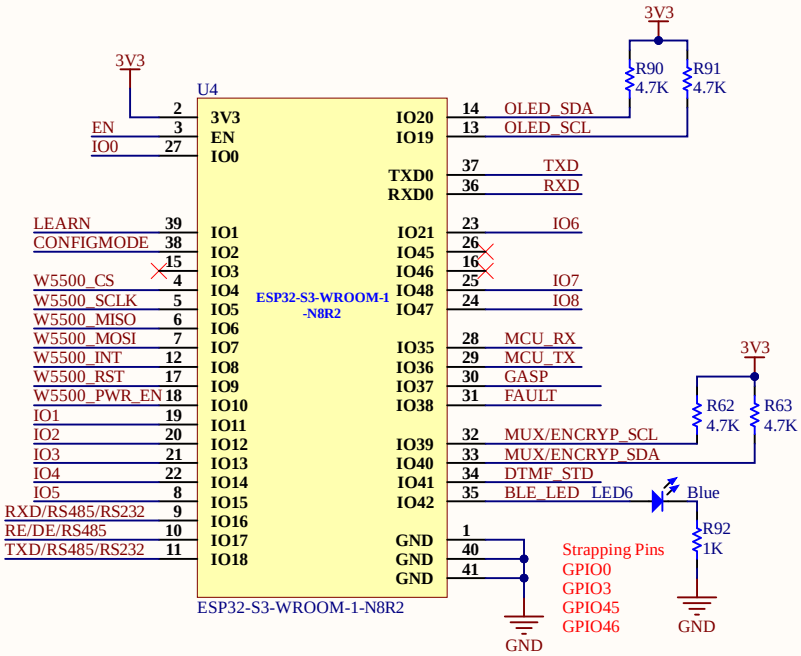
eSpark Consulting Group

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Date: 08/05/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 10

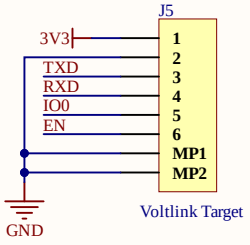
BUTTONS



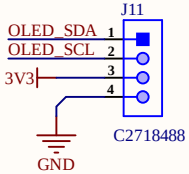
MICROCONTROLLER CIRCUIT ESP32-S3



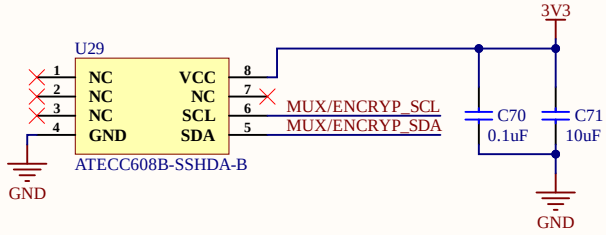
PROGRAMMING



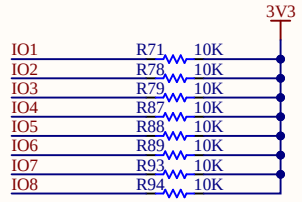
OLED HEADER

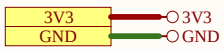


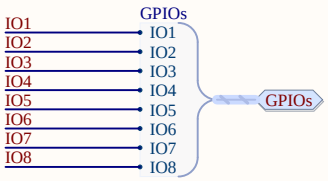
SECURE ELEMENT IC

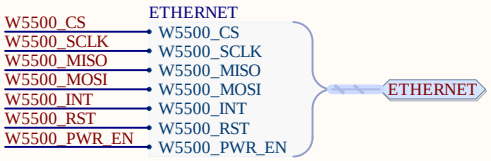


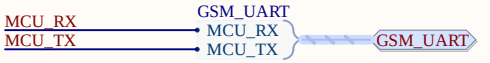
GPIO PULLUP



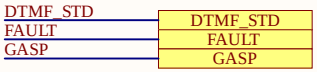










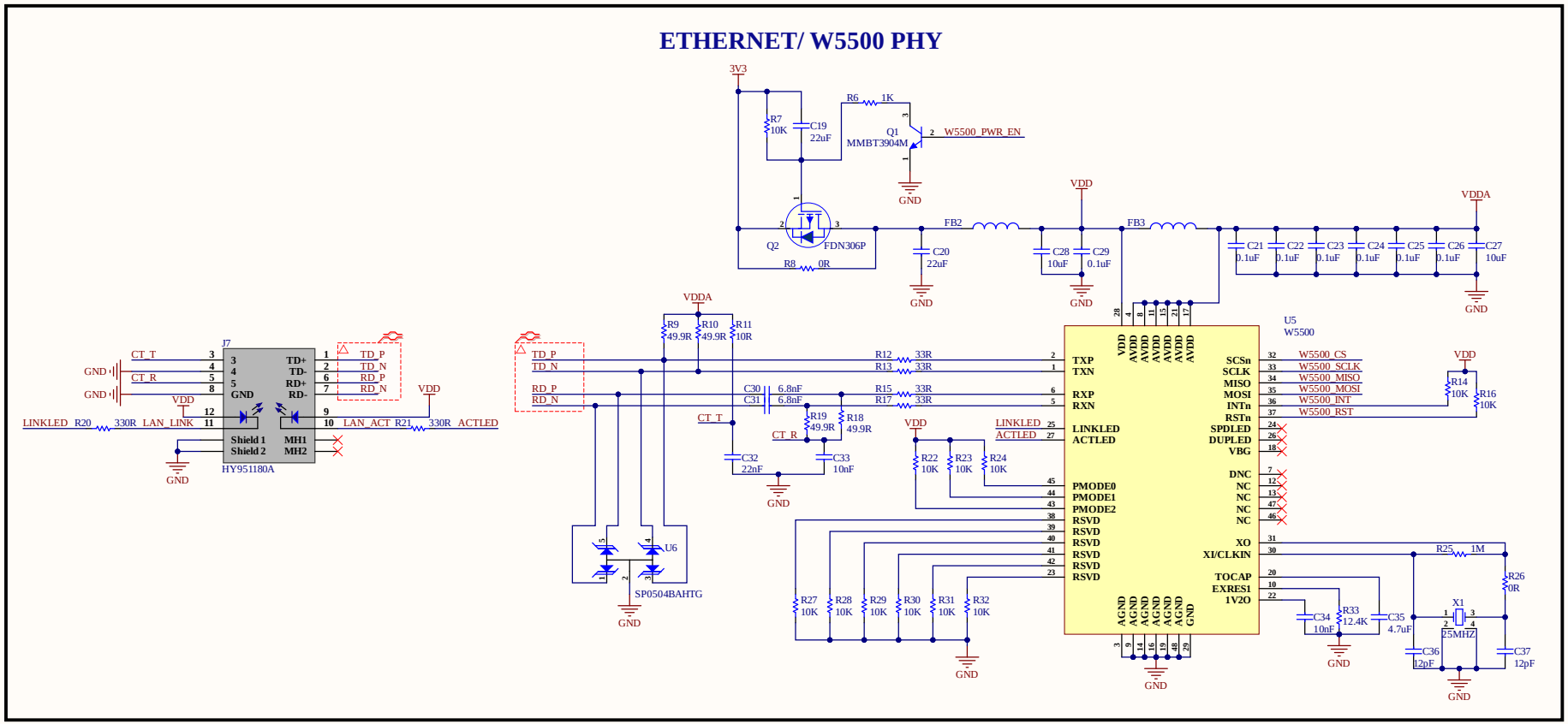




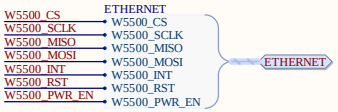
eSpark Consulting Group

Title: Controller_Sheet.SchDoc	
File: ESP32IOTProduct.PrjPcb	
Date: 08/05/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 10

ETHERNET/ W5500 PHY



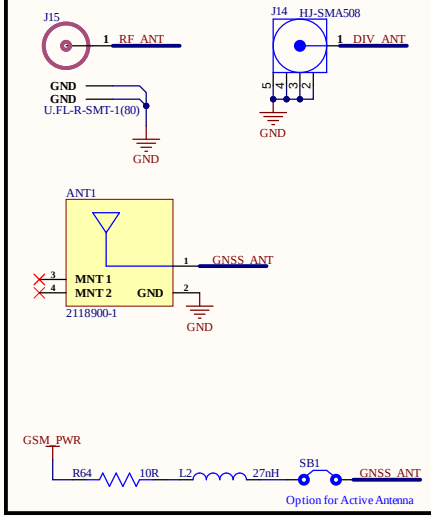




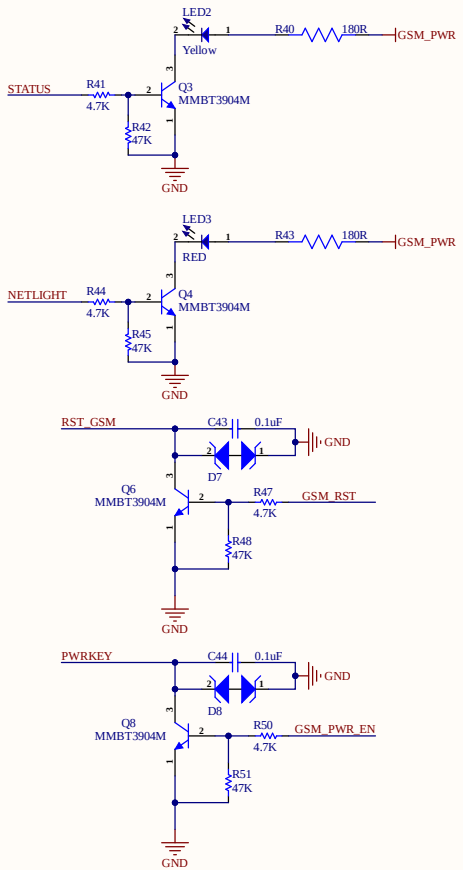
eSpark Consulting Group

Title: Ethernet_Sheet.SchDoc	
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Date: 08/05/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 10

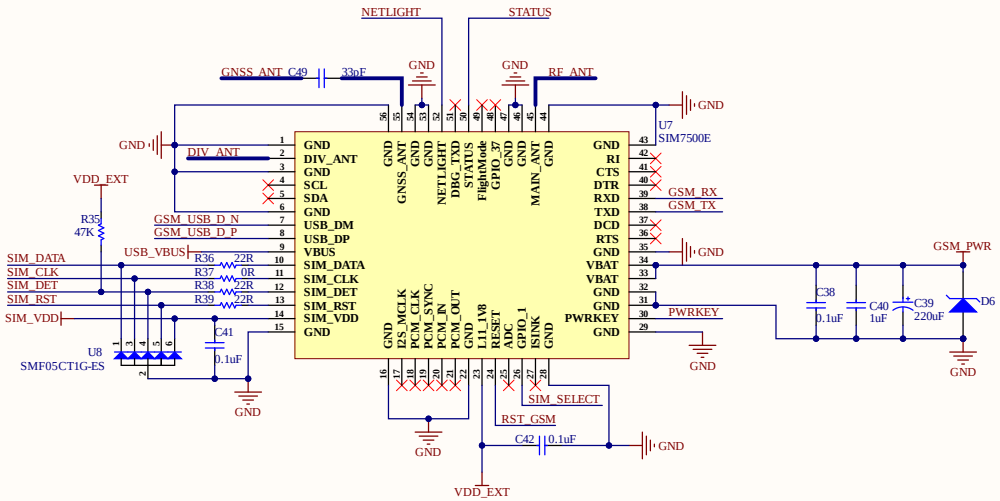
ANTENNA



GSM_COMMANDS



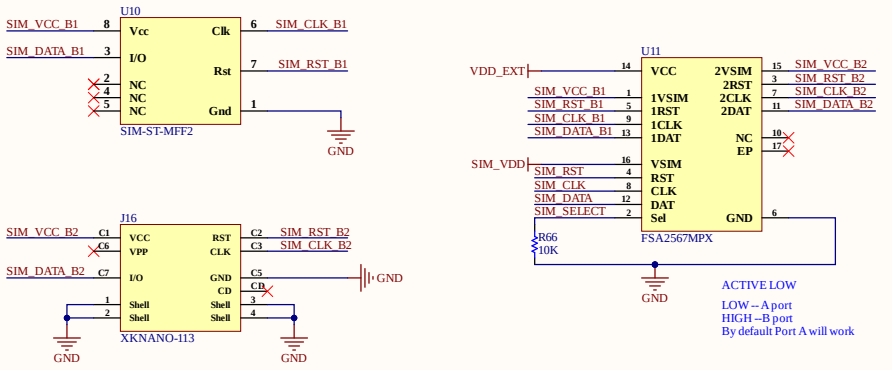
SIM7500E



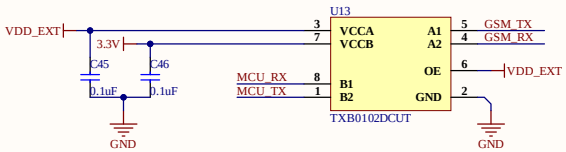
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

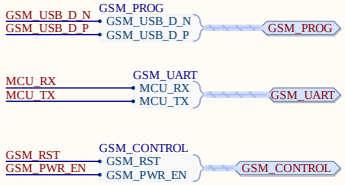
Datasheet Pg 31

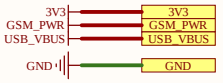
DUALSIM WITH MULTIPLEXER



Level Shifter for GSM Serial



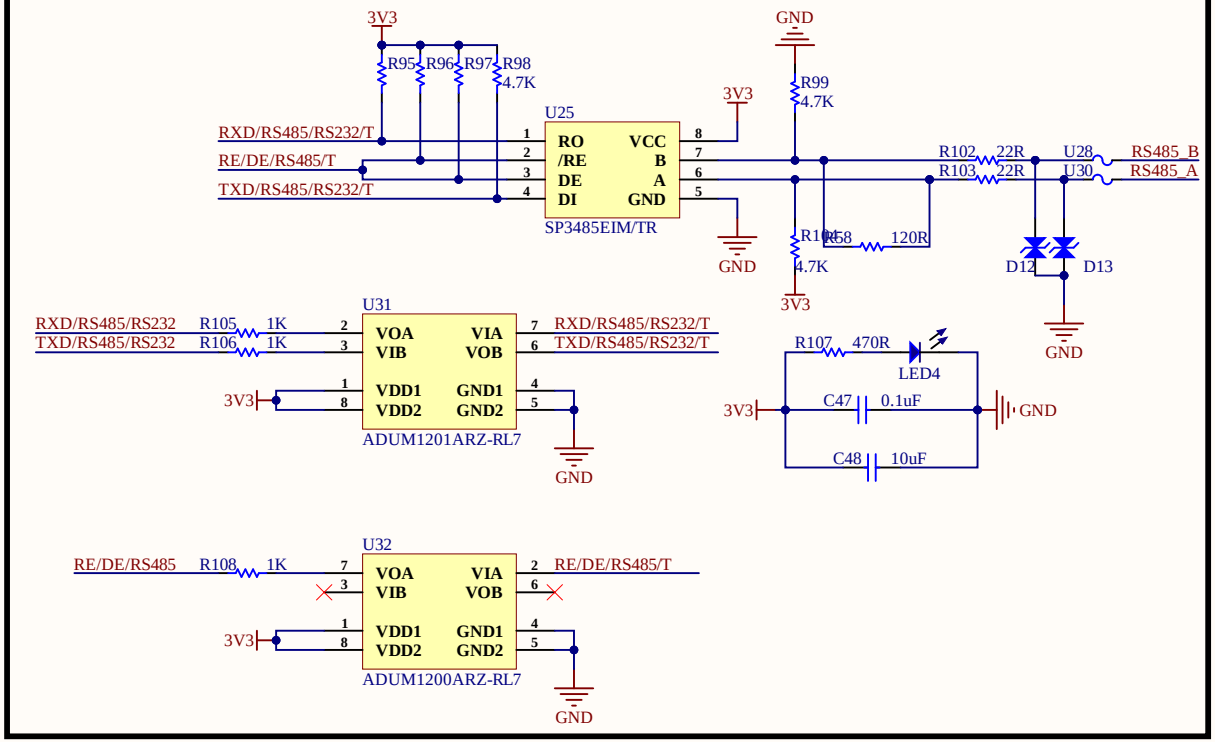




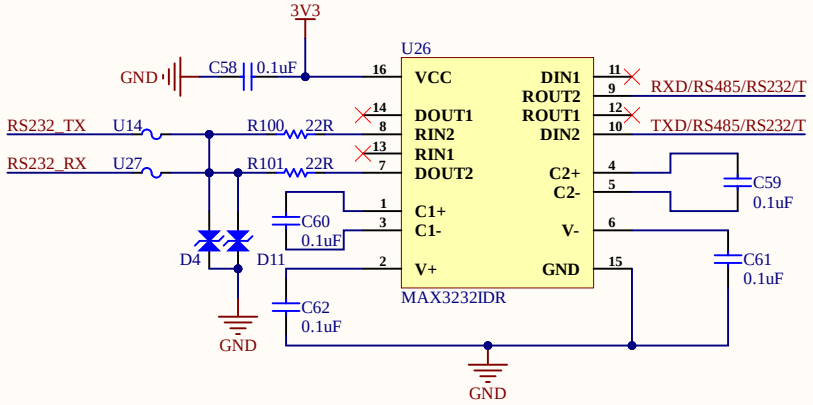
eSpark Consulting Group

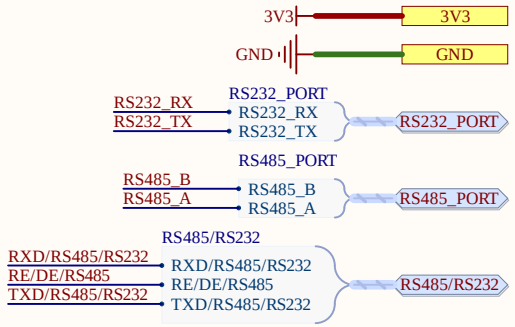
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Date: 08/05/2023	Page Num: 5
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RS485 to UART



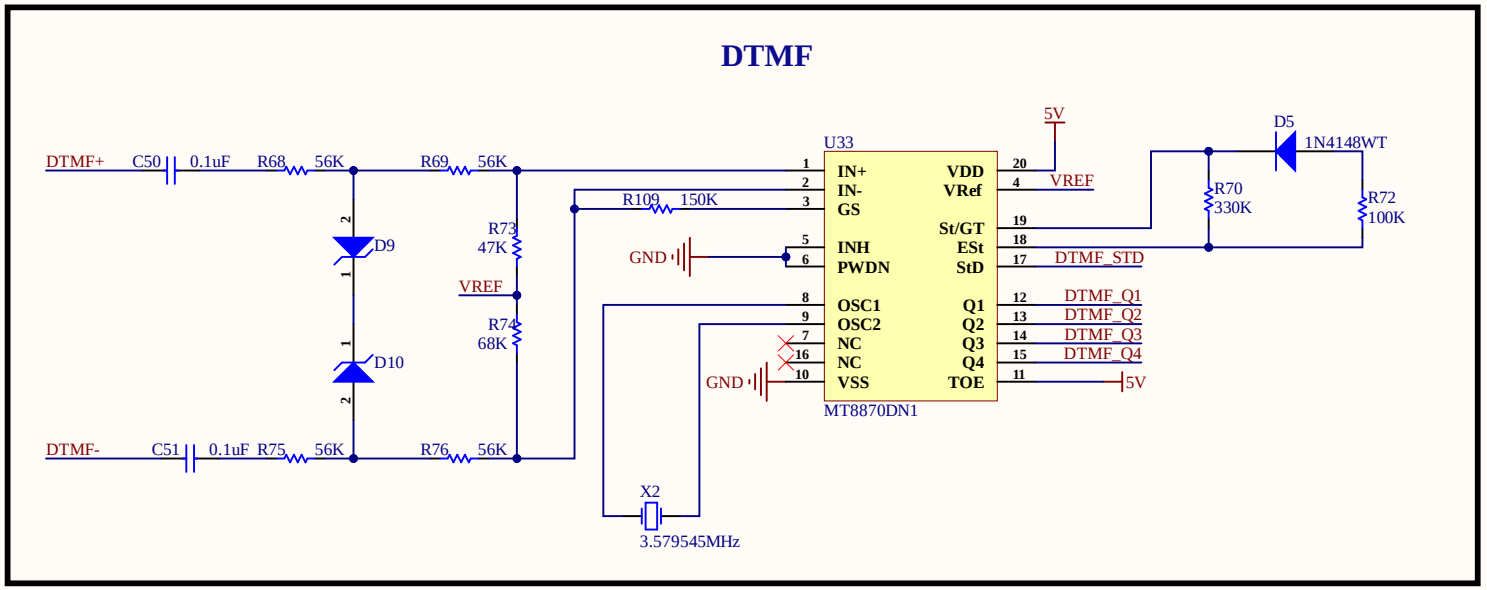
RS232 to UART

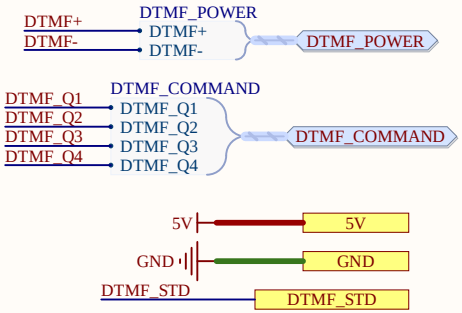




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Date: 08/05/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 10

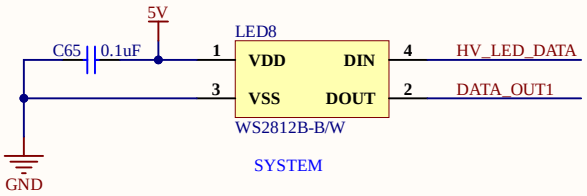




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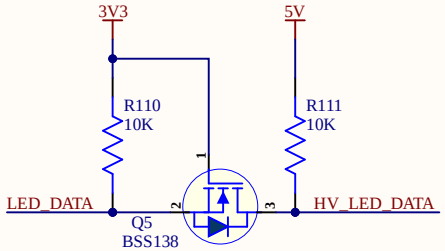
Title: DTMF_Decoder_Sheet.SchDoc	
File: ESP32IOTProduct.PrjPcb	
Date: 08/05/2023	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 10

SYSTEM LED



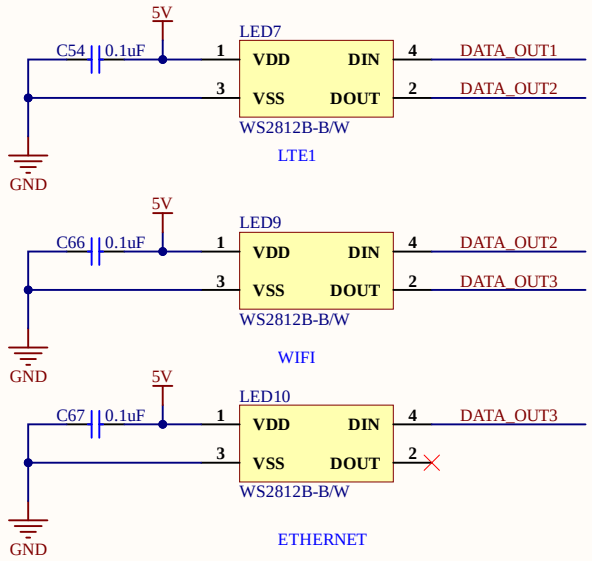
SYSTEM

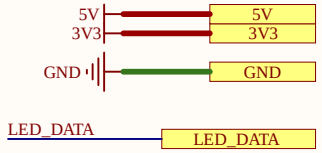
LEVEL SHIFTER



Bi-directional level shifter for WS2812B LED from 3.3V to 5V

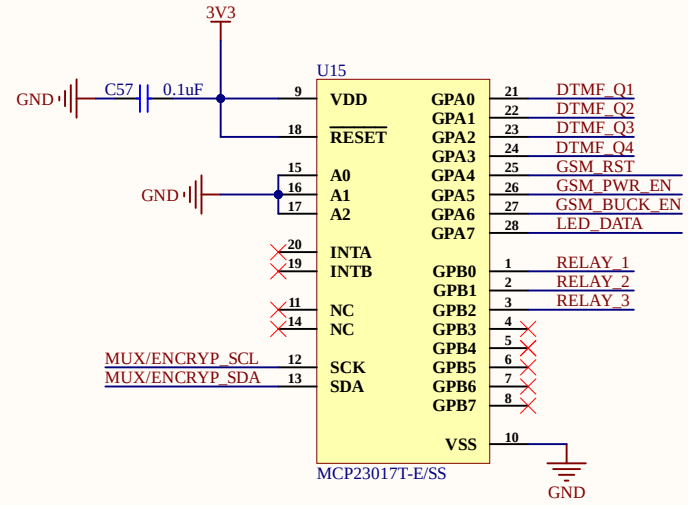
COMMS LED



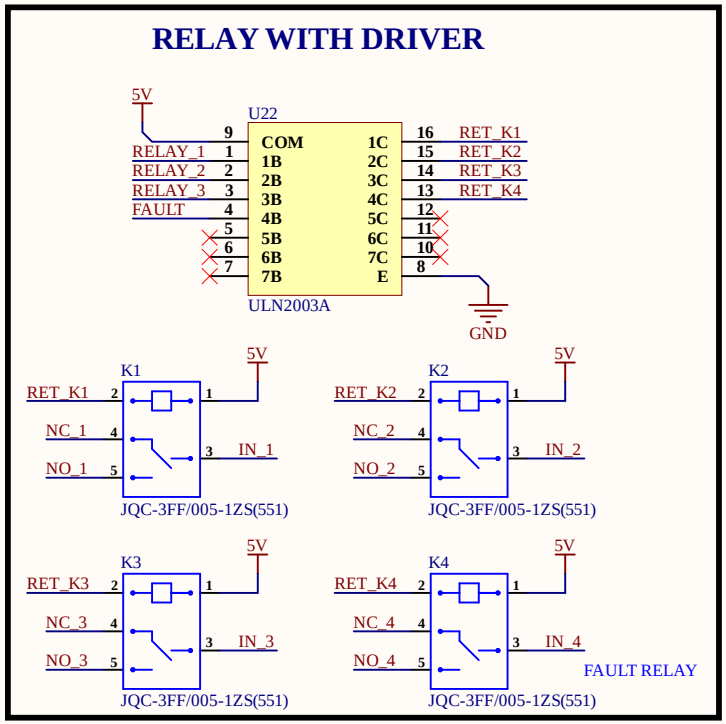


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Date: 08/05/2023	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 10

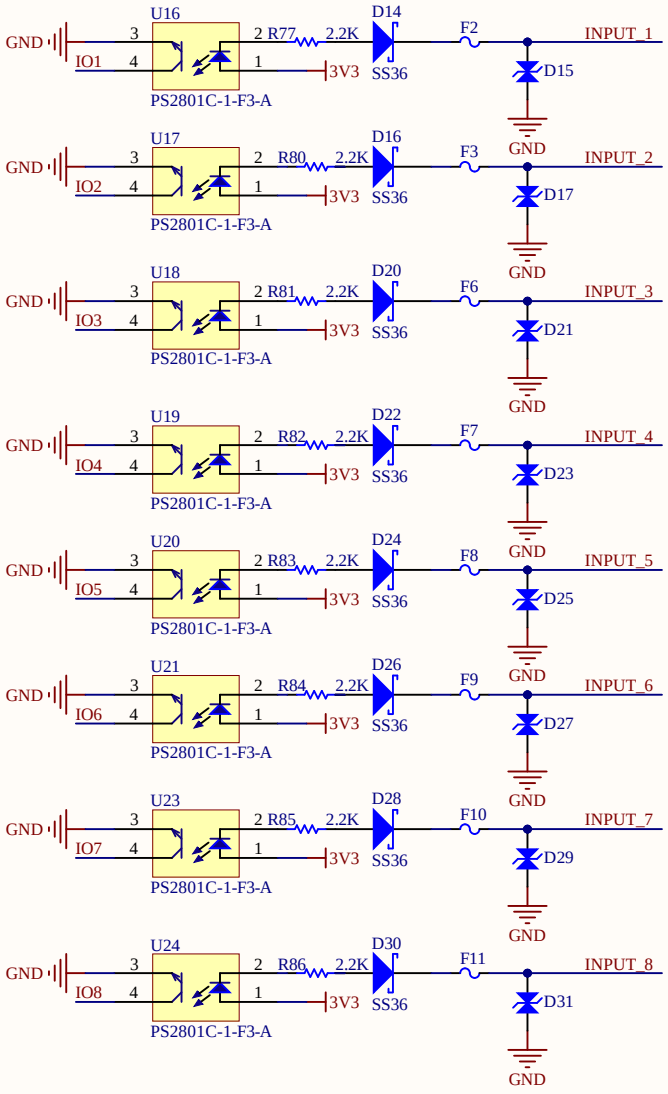
GPIO MULTIPLEXER



RELAY WITH DRIVER



OPTO ISOLATION FOR GPIOs



GSM_BUCK_EN
FAULT
LED_DATA

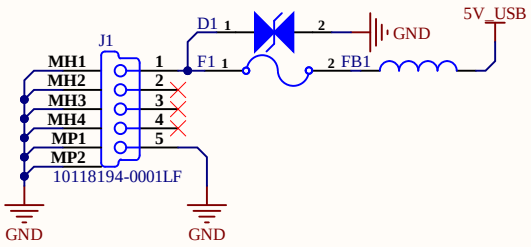
5V
3V3
GND

DTMF_COMMAND
GPIOs
9PIN_TERMINAL
GSM_CONTROL
MUX/ENCRYP
RELAY_PORT

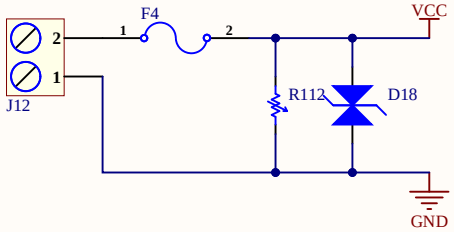
eSpark Consulting Group

Title: GPIOs_Expander_Relay_Sheet.SchDoc	
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Date: 08/05/2023	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 10

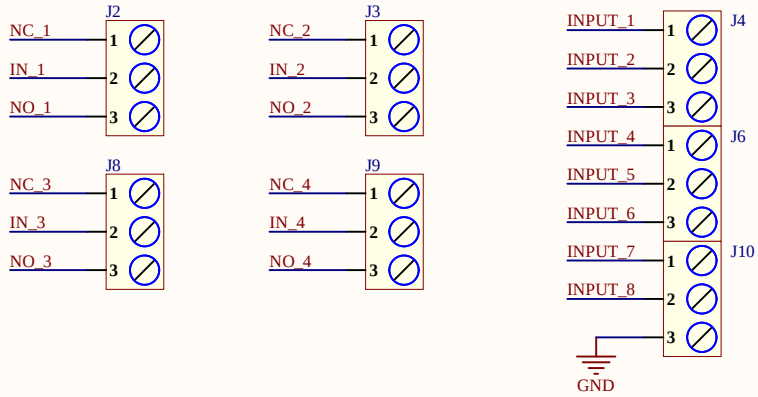
USB to Power Board



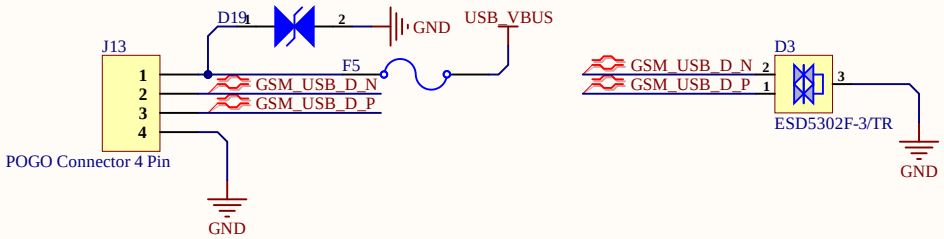
9-30V DC INPUT



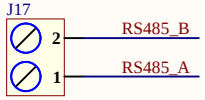
SCREW TERMINALS



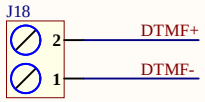
USB for GSM Updates



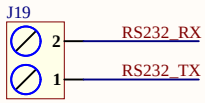
PORT for RS485

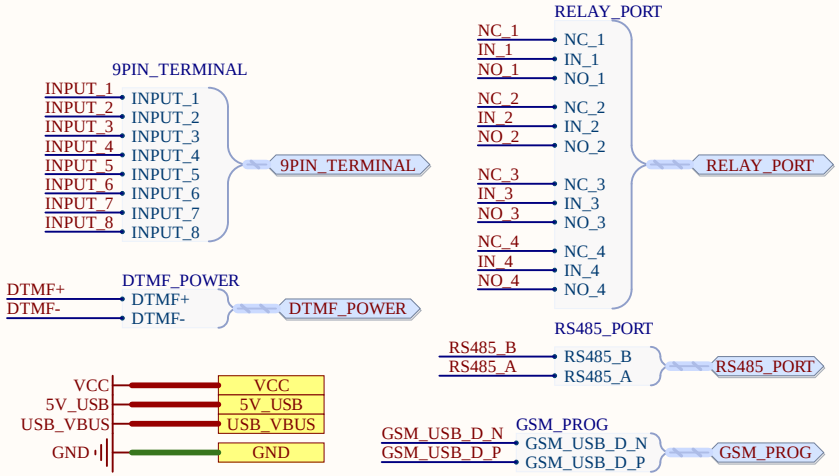


PORT for DTMF



PORT for RS232





Title: Connector_Sheet.SchDoc	
File: ESP32IOTProduct.PrjPcb	
Date: 08/05/2023	Page Num: 10
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